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24 Jan 2026 Shift 1 (Actual Paper)

Math Sec 1

1. If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$, then $\sin 7x\left(\cos \frac{13x}{2} + \sin \frac{13x}{2}\right) + \cos 7x\left(\cos \frac{13x}{2} - \sin \frac{13x}{2}\right)$ is equal to

(A) $\frac{1}{\sqrt{13}}$ (B) $\frac{5}{\sqrt{13}}$

(C) $\frac{6}{\sqrt{26}}$ (D) $\frac{4}{\sqrt{26}}$

2. Let $S = \frac{1}{25!} + \frac{1}{3!23!} + \frac{1}{5!21!} + \ldots$ up to 13 terms. If $13S = \frac{2^k}{n!}$, $k \in \mathrm{N}$, then $n + k$ is equal to

(A) 49 (B) 51 (C) 50 (D) 52

3. Let each of the two ellipses $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$ and $E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, (A < B)$ have eccentricity $\frac{4}{5}$. Let the lengths of the latus recta of E_1 and E_2 be l_1 and l_2, respectively, such that $2l_1^2 = 9l_2$. If the distance between the foci of E_1 is 8, then the distance between the foci of E_2 is

(A) $\frac{8}{5}$ (B) $\frac{32}{5}$

(C) $\frac{16}{5}$ (D) $\frac{96}{5}$

4. If the domain of the function
$f(x) = \log_{(10x^2-17x+7)}(18x^2 - 11x + 1)$ is
$(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$,
then $90(a + b + c + d + e)$ equals:

(A) 307 (B) 177 (C) 170 (D) 316

5. The value of $\frac{\sqrt{3}\,\mathrm{cosec}\,20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to

(A) 64 (B) 16 (C) 12 (D) 32

6. Let $A(1,0), B(2,-1)$ and $C\left(\frac{7}{3}, \frac{4}{3}\right)$ be three points. If the equation of the bisector of the angle ABC is $\alpha x + \beta y = 5$, then the value of $\alpha^2 + \beta^2$ is

(A) 5 (B) 10 (C) 13 (D) 8

7. Let $\alpha, \beta \in \mathrm{R}$ be such that the function

$$f(x) = \begin{cases} 2\alpha(x^2 - 2) + 2\beta x & , x < 1 \\ (\alpha + 3)x + (\alpha - \beta) & , x \geq 1 \end{cases}$$

be differentiable at all $x \in \mathrm{R}$. Then $34(\alpha + \beta)$ is equal to

(A) 24 (B) 84 (C) 36 (D) 48

8. Let $\mathrm{S} = \left\{z \in \mathrm{C} : \left|\frac{z-6i}{z-2i}\right| = 1 \text{ and } \left|\frac{z-8+2i}{z+2i}\right| = \frac{3}{5}\right\}$. Then $\sum_{z \in \mathrm{s}} |z|^2$ is equal to

(A) 413 (B) 398 (C) 423 (D) 385

9. Consider an A.P.: $a_1, a_2, \ldots, a_n$; $a_1 > 0$. If $a_2 - a_1 = \frac{-3}{4}$, $a_n = \frac{1}{4}a_1$, and
$\sum_{i=1}^{n} a_i = \frac{525}{2}$, then $\sum_{i=1}^{17} a_i$ is equal to

(A) 952 (B) 476 (C) 238 (D) 136

10. If the function $f(x) = \frac{e^x\left(e^{\tan x - x} - 1\right) + \log_e(\sec x + \tan x) - x}{\tan x - x}$ is co[n]tinuous at $x = 0$, then the value of $f(0)$ is equal to

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{3}{2}$ (D) $\frac{2}{3}$

11. From a lot containing 10 defective and 90 non-defecti[ve] bulbs, 8 bulbs are selected one by one with replacemen[t]. Then the probability of getting at least 7 defective bul[bs] is

(A) $\frac{7}{10^7}$ (B) $\frac{81}{10^8}$

(C) $\frac{67}{10^8}$ (D) $\frac{73}{10^8}$

12. The number of the real solutions of the equation $x|x - 3| + |x - 1| - 2 = 0$ is

(A) 4 (B) 2 (C) 3 (D) 5

13. Let the lines $\mathrm{L}_1 : \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} - 4\hat{k}), \lambda \in \mathrm{R}$ and $\mathrm{L}_2 : \vec{r} = (4\hat{i} + \hat{j}) + \mu(5\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathrm{R}$, intersect at the point R. Let P and Q be the points lyi[ng] on lines L_1 and L_2, respectively, such that $|\overrightarrow{\mathrm{PR}}| = \sqrt{2}$[...] and $|\overrightarrow{\mathrm{PQ}}| = \sqrt{\frac{47}{3}}$. If the point P lies in the first octan[t] then $27(\mathrm{QR})^2$ is equal to

(A) 320 (B) 340 (C) 360 (D) 348

14. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \vec{a} \times \vec{b}$. Let [...] be a vector such that $|\vec{d} - \vec{a}| = \sqrt{11}$, $|\vec{c} \times \vec{d}| = 3$ a[nd] the angle between $\vec{c}$ and $\vec{d}$ is $\frac{\pi}{4}$. Then $\vec{a} \cdot \vec{d}$ is equal to

(A) 1 (B) 0 (C) 11 (D) 3

15. Let $729, 81, 9, 1, \ldots$ be a sequence and P_n denote th[e] product of the first n terms of this sequence.
If $2\sum_{n=1}^{40}(\mathrm{P}_n)^{\frac{1}{n}} = \frac{3^\alpha - 1}{3^\beta}$ and $\gcd(\alpha, \beta) = 1$, then $\alpha + \beta$ is equal to

(A) 76 (B) 73 (C) 74 (D) 75

16. Let R be a relation defined on the set $\{1, 2, 3, 4\} \times$ $\{1, 2, 3, 4\}$ by $R = \{((a, b), (c, d)) : 2a + 3b = 3c + 4$[...]. Then the number of elements in R is

(A) 6 (B) 15 (C) 18 (D) 12

17. Let A_1 be the bounded area enclosed by the curves $y =$ $x^2 + 2$, $x + y = 8$ and y-axis that lies in the first qua[d]rant. Let A_2 be the bounded area enclosed by the curv[es] $y = x^2 + 2$, $y^2 = x$, $x = 2$, and y-axis that lies in the fir[st] quadrant. Then $A_1 - A_2$ is equal to

(A) $\frac{2}{3}(\sqrt{2} + 1)$ (B) $\frac{2}{3}(3\sqrt{2} + 1)$

(C) $\frac{2}{3}(4\sqrt{2} + 1)$ (D) $\frac{2}{3}(2\sqrt{2} + 1)$

18. Let $f(t)=\int\left(\frac{1-\sin(\log_e t)}{1-\cos(\log_e t)}\right)dt$, for $t>1$.

If $f\left(e^{\pi/2}\right)=-e^{\pi/2}$ and $f\left(e^{\pi/4}\right)=\alpha e^{\pi/4}$, then find the value of α.

(A) $-1-2\sqrt{2}$ (B) $-1+\sqrt{2}$

(C) $1+\sqrt{2}$ (D) $-1-\sqrt{2}$

19. Let a circle of radius 4 pass through the origin O, the points $A(-\sqrt{3}a,0)$ and $B(0,-\sqrt{2}b)$, where a and b are real parameters and $ab\neq 0$. Then the locus of the centroid of $\triangle OAB$ is a circle of radius

(A) $\frac{7}{3}$ (B) $\frac{5}{3}$

(C) $\frac{8}{3}$ (D) $\frac{11}{3}$

20. The mean and variance of a data of 10 observations ar 10 and 2, respectively. If an observations α in this data replaced by β, then the mean and variance become 10. and 1.99 , respectively. Then $\alpha+\beta$ equals

(A) 15 (B) 5 (C) 20 (D) 10

Math Sec 2

21. Let a differentiable function f satisfy the equation $\int_0^{36} f\left(\frac{tx}{36}\right)dt=4\alpha f(x)$. If $y=f(x)$ is a standard parabola passing through the points $(2,1)$ and $(-4,\beta)$, then β^α is equal to ____ .

22. Let $(2a,a)$ be the largest interval in which the function $f(t)=\frac{|t+1|}{t^2}, t<0$, is strictly decreasing. Then the local maximum value of the function $g(x)=2\log_e(x-2)+ax^2+4x-a, x>2$, is ____

23. Let a line L passing through the point $\mathrm{P}(1,1,1)$ be pe pendicular to the lines $\frac{x-4}{4}=\frac{y-1}{1}=\frac{z-1}{1}$ and $\frac{x-17}{1}=\frac{y-71}{1}=\frac{z}{0}$. Let the line L intersect the yz plane at the poi Q. Another line parallel to L and passing through th point $\mathrm{S}(1,0,-1)$ intersects the yz-plane at the point I Then the square of the area of the parallelogram PQRS equal to ____ .

24. The number of numbers greater than 5000 , less tha 9000 and divisible by 3 , that can be formed using th digits $0,1,2,5,9$, if the repetition of the digits is allowe is ____

25. The number of 3×2 matrices A, which can be forme using the elements of the set $\{-2,-1,0,1,2\}$ such th the sum of all the diagonal elements of A^TA is 5, is

Phy Sec 1

26. A boy throws a ball into air at 45° from the horizontal to land it on a roof of a building of height H. If the ball attains maximum height in 2 s and lands on the building in 3 s after launch, then value of H is ______ m. $(g=10\ \mathrm{m/s^2})$

(A) 15 (B) 10 (C) 25 (D) 20

27. The electrostatic potential in a charged spherical region of radius r varies as $V=ar^3+b$, where a and b are constants. The total charge in the sphere of unit radius is $\alpha\times\pi a\epsilon_0$. The value of α is ____ .(permittivity of vacuum is ϵ_0)

(A) -9 (B) -6 (C) -12 (D) -8

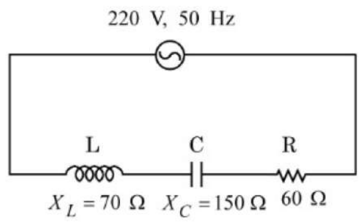
28. Two electrons are moving in orbits of two hydrogen like atoms with speeds $3\times 10^5\ \mathrm{m/s}$ and $2.5\times 10^5\ \mathrm{m/s}$ respectively. If the radii of these orbits are nearly same then the possible order of energy states are ______ respectively.

(A) 8 and 10 (B) 9 and 8 (C) 10 and 12 (D) 6 and 5

29. The exit surface of a prism with refractive index n coated with a material having refractive index $\frac{n}{2}$. Whe this prism is set for minimum angle of deviation, it ex actly meets the condition of critical angle. The prism an gle is ____ .

(A) 60° (B) 45° (C) 15° (D) 30°

30. For the series LCR circuit connected with 220 V, 50 H a.c source as shown in the figure, the power factor is $\frac{\alpha}{1}$ The value of α is ____ .



(A) 8 (B) 10 (C) 4 (D) 6

31. Match the following lists:

	List-I		List-II